

ADAMAS UNIVERSITY

Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION

FOR

ONLINE EXAMINATION SYSTEM

Version 1.0

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for the Online Examination System, Version 1.0. The system is designed to conduct examinations through an online platform and to automate major examination activities including examination creation, question management, examination participation, answer submission, answer evaluation, and result generation.

This SRS specifies the functional and non-functional requirements of the Online Examination System. It covers the requirements of students and administrators who interact with the system.

The document provides a common reference for the development, testing, evaluation, and future maintenance of the system.

1.2 Document Conventions

The following conventions are used throughout this SRS:

- SRS refers to Software Requirements Specification.
- OES refers to Online Examination System.
- FR refers to Functional Requirement.
- NFR refers to Non-Functional Requirement.
- REQ-XX is used to uniquely identify individual functional requirements.
- PERF-XX is used to identify performance requirements.
- SEC-XX is used to identify security requirements.
- SAFE-XX is used to identify safety and data-protection requirements.
- BR-XX is used to identify business rules.
- TBD-XX is used to identify requirements that remain to be determined.
- Requirement priorities are classified as High, Medium, or Low.
- The term “shall” is used to specify a mandatory system requirement.
- Section headings are numbered hierarchically, such as 1.1, 1.2, 2.1, etc.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for developers, project managers, administrators, testers, teachers or examination coordinators, and students who are the primary users of the Online Examination System.

Developers should refer to the functional requirements, external interface requirements, and non-functional requirements to understand the expected behavior and constraints of the system.

Testers should focus on the system features, stimulus/response sequences, functional requirements, performance requirements, security requirements, and business rules to prepare appropriate test cases.

Administrators and examination coordinators should review the overall description and system features to understand the functions available for examination management.

Students should review the product scope, user interface requirements, and examination-related system features to understand how examinations are conducted.

The document should preferably be read in the following order:

1. Introduction
2. Overall Description
3. External Interface Requirements
4. System Features
5. Other Nonfunctional Requirements
6. Other Requirements
7. Appendices

1.4 Product Scope

The Online Examination System is a web-based software application designed to automate the major operations involved in conducting online examinations.

The system will provide facilities for administrator authentication, question management, examination creation, examination scheduling, student examination participation, answer submission, automatic evaluation of objective-type answers, and result generation.

Students will be able to log in, view available examinations, read examination instructions, attempt questions, submit answers, and view their results when released.

Administrators will be able to create and manage examinations, add and update questions, define correct answers and marks, schedule examinations, and view examination results.

The major objectives of the system are to reduce manual work, improve the accuracy of answer evaluation, reduce the time required for result generation, provide centralized examination management, and offer students a convenient mechanism for taking examinations.

The system does not cover physical examination infrastructure, physical invigilation, or advanced proctoring unless such functionality is explicitly added in a future version.

1.5 References

1. Karl E. Wiegers, *Software Requirements Specification Template*, Software Requirements Engineering.
2. IEEE, *IEEE Recommended Practice for Software Requirements Specifications*.
3. Object Management Group, *Unified Modeling Language (UML) Specification*.
4. MySQL 8.0 Reference Manual.
5. Project-specific requirements for the Online Examination System.

2 Overall Description

2.1 Product Perspective

The Online Examination System is a new web-based application intended to replace or reduce manual examination activities. The system will provide a centralized platform for managing questions and examinations and for evaluating objective-type answers.

Students will interact with the system through a web browser to access scheduled examinations, answer questions, submit their responses, and view results.

Administrators will use the system to manage question banks, create examinations, schedule examinations, configure marks and correct answers, and access results.

The system will store and retrieve user, examination, question, answer, and result information using a centralized database.

The high-level context of the system is represented below.

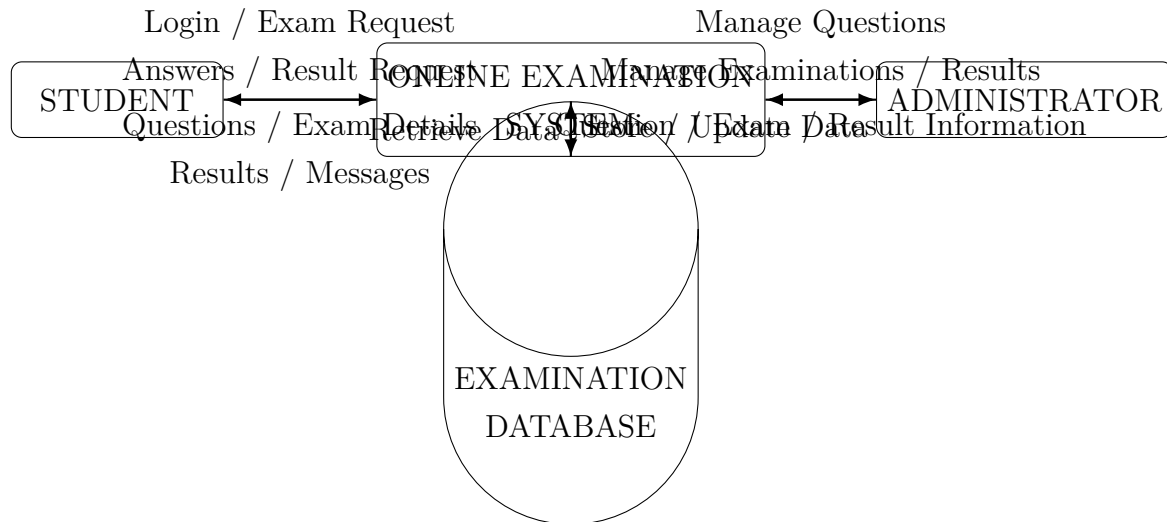


Figure 1: High-level context of the Online Examination System

2.2 Product Functions

The major functions of the Online Examination System include:

- User registration and authentication.
- Student account management.
- Question addition, modification, deletion, and organization.
- Examination creation and configuration.
- Examination scheduling.
- Assignment of questions to examinations.
- Viewing available examinations by students.
- Conducting timed online examinations.
- Recording and submitting student answers.
- Automatic evaluation of objective-type answers.
- Calculation of examination scores.
- Generation and storage of examination results.
- Viewing examination results.
- Administrative access to examination and result information.

2.3 User Classes and Characteristics

The Online Examination System supports the following major user classes.

- **Student:** Students are regular users with basic computer knowledge. They can log in, view available examinations, attempt examinations, submit answers, and view their results. Their access level is limited to examination participation and their own result information.
- **Administrator:** Administrators are authorized users responsible for managing examinations and questions. They can create and manage examinations, manage questions, schedule examinations, and view results. Their access level is high.

Table 1: User Classes and Their Characteristics

User Class	Main Functions	Access Level
Student	View exams, attempt exams, submit answers, view results	Limited
Administrator	Manage users, questions, examinations, schedules, and results	High

2.4 Operating Environment

The Online Examination System will operate as a web-based application. Users will access the system through modern web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox.

The application server may operate on a Windows or Linux-based server. MySQL may be used as the database management system.

The expected operating environment includes:

- Hardware: Desktop computers, laptops, tablets, and application server.
- Operating System: Windows or Linux for the server; standard supported operating systems for clients.
- Web Browser: Google Chrome, Microsoft Edge, Mozilla Firefox, or equivalent modern browser.
- Database: MySQL.
- Network: University local network or Internet connection.

2.5 Design and Implementation Constraints

The following constraints apply to the development of the Online Examination System:

- The system shall be developed as a web-based application.
- MySQL shall be used as the database management system unless changed by the project implementation plan.
- The system shall use secure communication protocols for transmitting sensitive information.
- Administrative functions shall be accessible only to authorized users.
- User passwords shall not be stored in plain text.
- Students shall not be allowed to access correct answers before evaluation.
- Examination access shall be controlled according to the configured schedule.
- The system shall be compatible with standard university computers and web browsers.

2.6 User Documentation

The system will be accompanied by user documentation to assist different categories of users.

The following documentation will be provided:

- **Student User Manual:** Explains login, viewing available examinations, starting an examination, answering questions, submitting an examination, and viewing results.
- **Administrator User Manual:** Explains question management, examination creation, scheduling, result management, and administrative functions.
- **Online Help:** Provides brief explanations and guidance for commonly used system functions.

2.7 Assumptions and Dependencies

Assumptions

- Each student and administrator has a unique identification or login credential.

- Users have basic computer and web browsing knowledge.
- Administrators will maintain accurate question and examination information.
- Administrators will enter correct answers and marks correctly.
- Students will have access to a suitable device and network connection during examinations.
- The examination rules and marking scheme will be defined before the examination begins.

Dependencies

- The system depends on the database server for storing and retrieving examination information.
- The system depends on the web/application server for providing access to users.
- The system depends on network connectivity for communication between client browsers and the application server.
- The system depends on the availability of a modern web browser on the client device.

3 External Interface Requirements

This section describes the interfaces between the Online Examination System and its users, hardware devices, software components, and communication networks.

3.1 User Interfaces

The Online Examination System will provide a web-based graphical user interface for students and administrators. The interface will be designed to be simple, consistent, and easy to use.

The major user interface components are described below:

- **Login Interface:** The system shall provide username/email and password fields along with a login button. Invalid login credentials shall generate an appropriate error message.

- **Student Dashboard:** The student dashboard shall provide options for viewing available examinations, examination instructions, examination attempts, submissions, and results.
- **Administrator Dashboard:** The administrator dashboard shall provide options for managing questions, examinations, schedules, users, and results.
- **Question Management Interface:** The administrator shall be provided with forms to add, edit, delete, and view questions and their answer options.
- **Examination Interface:** The examination screen shall display questions, answer options, navigation controls, submission controls, and the remaining examination time.
- **Result Interface:** The system shall display relevant result information such as examination name, marks obtained, total marks, percentage, and result status.
- **Navigation:** The system shall provide clearly labelled navigation menus and buttons. A logout option shall be available to authenticated users.
- **Error Messages:** The system shall display meaningful error messages when users enter invalid or incomplete information.

The user interface shall maintain a consistent layout, terminology, button placement, and navigation structure throughout the system.

3.2 Hardware Interfaces

The Online Examination System will primarily operate on standard desktop computers, laptops, tablets, and application servers.

The following hardware interfaces may be supported:

- **Client Computer:** Students and administrators will access the system using desktop computers, laptops, or compatible devices.
- **Server:** The application server will host the Online Examination System and provide access to users through the network.
- **Input Devices:** The system shall be compatible with standard keyboard, mouse, and touchscreen input where supported by the browser.
- **Camera or Microphone:** These are not required for the baseline system. If online proctoring is introduced in a future version, appropriate interfaces may be added.

3.3 Software Interfaces

The Online Examination System will interact with several software components required for its operation.

Table 2: Software Interfaces

Software	Version	Purpose
MySQL	8.0 or compatible	Database management
Web Browser	Current supported version	User access
Operating System	Windows/Linux	System operation
Web/Application Server	Project-selected version	Application hosting

The system will use the database to store and retrieve information related to users, examinations, questions, answer options, submitted answers, and results.

The major data exchanged between the application and the database includes:

- User information.
- Examination information.
- Question and option information.
- Correct answer information.
- Student answer submissions.
- Evaluation information.
- Result information.

The application will send requests to the database to create, retrieve, update, and delete records. The database will return the requested information to the application.

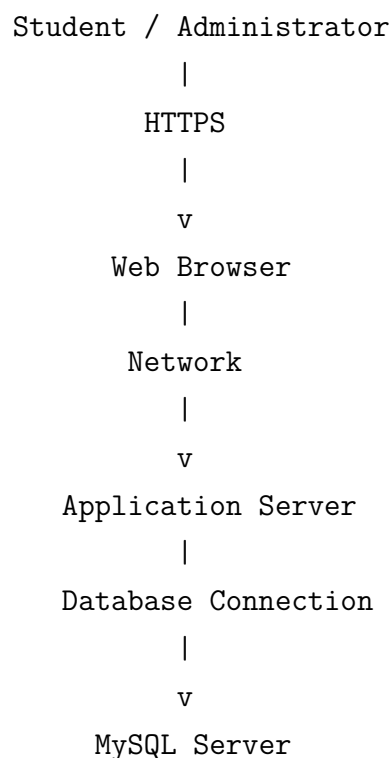
3.4 Communications Interfaces

The Online Examination System will use network communication to allow users to access the web-based application.

The following communication requirements apply:

- The system shall use HTTP/HTTPS protocols for communication between the client and the application server.
- HTTPS shall be used when transmitting sensitive information such as user credentials and examination data.
- The system shall support communication between client browsers and the application server through a standard Internet or local network connection.
- The system shall provide appropriate error handling when network communication is interrupted.
- Data transmitted between the client and server shall follow the required application data format.

The general communication architecture can be represented as:



4 System Features

This section describes the major functional features of the Online Examination System. Each feature includes its description and priority, stimulus/response sequences, and de-

tailed functional requirements.

4.1 User Login

4.1.1 Description and Priority

The User Login feature allows students and administrators to securely access the Online Examination System using their registered credentials.

Priority: High

4.1.2 Stimulus/Response Sequences

Table 3: Stimulus/Response Sequence for User Login

Stimulus / User Action	System Response
User enters username/email and password	System accepts and validates the input.
User clicks Login	System validates the entered credentials.
Valid credentials are provided	System opens the appropriate user dashboard.
Invalid credentials are provided	System displays an appropriate error message.
Username or password is empty	System asks the user to provide the required information.

4.1.3 Functional Requirements

- **REQ-01:** The system shall allow registered users to enter their username/email and password.
- **REQ-02:** The system shall validate the entered username/email and password.
- **REQ-03:** The system shall authenticate the user when valid credentials are provided.
- **REQ-04:** The system shall redirect authenticated students to the student dashboard.

- **REQ-05:** The system shall redirect authenticated administrators to the administrator dashboard.
- **REQ-06:** The system shall display an error message when invalid credentials are entered.
- **REQ-07:** The system shall display an appropriate validation message when the login fields are empty.
- **REQ-08:** The system shall prevent unauthenticated users from accessing protected pages.

4.2 Question Management

4.2.1 Description and Priority

The Question Management feature allows authorized administrators to create, view, update, and delete examination questions. Questions may contain answer options, correct answers, marks, and other relevant information.

Priority: High

4.2.2 Stimulus/Response Sequences

Table 4: Stimulus/Response Sequence for Question Management

Stimulus / User Action	System Response
Administrator selects Add Question	System displays the question entry form.
Administrator enters question details	System validates the entered information.
Administrator clicks Save	System stores the question in the database.
Administrator selects Update Question	System displays existing question information.
Administrator submits updated information	System validates and updates the question record.
Administrator selects Delete Question	System verifies authorization and removes or deactivates the question.

4.2.3 Functional Requirements

- **REQ-09:** The system shall allow authorized administrators to add new questions.
- **REQ-10:** The system shall allow authorized administrators to define answer options for supported objective questions.
- **REQ-11:** The system shall allow authorized administrators to define the correct answer.
- **REQ-12:** The system shall allow authorized administrators to assign marks to a question.
- **REQ-13:** The system shall allow authorized administrators to update question information.
- **REQ-14:** The system shall allow authorized administrators to remove or deactivate questions.
- **REQ-15:** The system shall validate required question information before storing a question.

- **REQ-16:** The system shall prevent students from modifying or accessing correct answer information.

4.3 Examination Management

4.3.1 Description and Priority

The Examination Management feature allows administrators to create, configure, update, schedule, and manage examinations.

Priority: High

4.3.2 Stimulus/Response Sequences

Table 5: Stimulus/Response Sequence for Examination Management

Stimulus / User Action	System Response
Administrator selects Create Examination	System displays the examination creation form.
Administrator enters examination details	System validates the entered information.
Administrator selects questions	System associates selected questions with the examination.
Administrator saves examination	System stores the examination information.
Administrator schedules examination	System records the configured start and end time.
Administrator updates examination	System validates and stores the updated information.

4.3.3 Functional Requirements

- **REQ-17:** The system shall allow authorized administrators to create an examination.
- **REQ-18:** The system shall allow administrators to specify the examination title and description.

- **REQ-19:** The system shall allow administrators to define the examination duration.
- **REQ-20:** The system shall allow administrators to define the examination start and end time.
- **REQ-21:** The system shall allow administrators to assign questions to an examination.
- **REQ-22:** The system shall allow administrators to configure the total marks of an examination.
- **REQ-23:** The system shall allow administrators to update examination information before the examination becomes active.
- **REQ-24:** The system shall prevent unauthorized users from creating or modifying examinations.
- **REQ-25:** The system shall prevent students from accessing an examination outside its permitted schedule.

4.4 Conduct Examination

4.4.1 Description and Priority

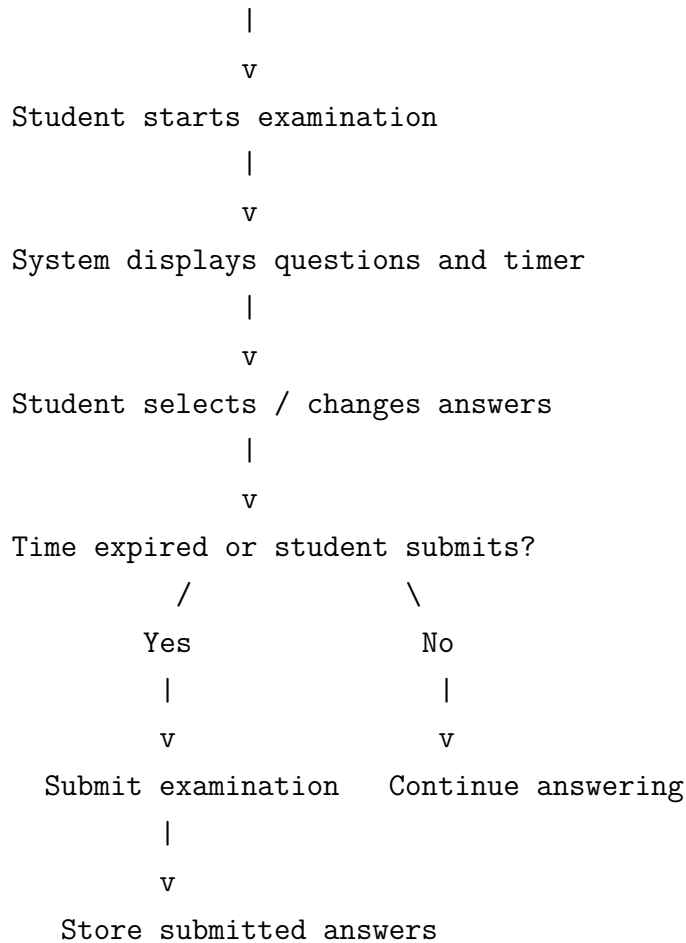
The Conduct Examination feature allows eligible students to participate in scheduled examinations, answer questions, navigate through the examination, and submit their responses within the permitted duration.

Priority: High

4.4.2 Stimulus/Response Sequences

The examination process can be represented as follows:

```
Student selects an available examination
      |
      v
System verifies examination schedule
      |
      v
System displays examination instructions
```



4.4.3 Functional Requirements

- **REQ-26:** The system shall display examinations available to the authenticated student.
- **REQ-27:** The system shall verify that the student is eligible to access the selected examination.
- **REQ-28:** The system shall display examination instructions before the examination begins.
- **REQ-29:** The system shall allow an eligible student to start an examination during its permitted time.
- **REQ-30:** The system shall display the questions and available answer options.
- **REQ-31:** The system shall display the remaining examination time.
- **REQ-32:** The system shall allow students to select and change answers before final submission.

- **REQ-33:** The system shall record student answers during the examination.
- **REQ-34:** The system shall prevent students from starting an examination after its permitted access period.
- **REQ-35:** The system shall automatically submit or lock the examination when the configured duration expires.

4.5 Answer Evaluation

4.5.1 Description and Priority

The Answer Evaluation feature automatically evaluates supported objective-type answers submitted by students and calculates the corresponding marks.

Priority: High

4.5.2 Stimulus/Response Sequences

```
Student submits examination
      |
      v
System retrieves submitted answers
      |
      v
System retrieves correct answers
      |
      v
Compare submitted and correct answers
      |
      v
Calculate marks for each question
      |
      v
Calculate total score
      |
      v
Store evaluation information
```

4.5.3 Functional Requirements

- **REQ-36:** The system shall retrieve the answers submitted by the student.
- **REQ-37:** The system shall retrieve the corresponding correct answers for objective-type questions.
- **REQ-38:** The system shall compare submitted answers with the configured correct answers.
- **REQ-39:** The system shall award marks according to the configured marking scheme.
- **REQ-40:** The system shall assign zero marks to unanswered questions unless another rule has been configured.
- **REQ-41:** The system shall calculate the student's total marks.
- **REQ-42:** The system shall store evaluation information for result generation.
- **REQ-43:** The system shall not allow students to modify evaluated answers or marks.

4.6 Result Generation and Management

4.6.1 Description and Priority

The Result Generation and Management feature generates examination results after evaluation and allows authorized users to access the appropriate result information.

Priority: High

4.6.2 Stimulus/Response Sequences

Table 6: Stimulus/Response Sequence for Result Generation

Stimulus / User Action	System Response
Examination evaluation is completed	System calculates the final score.
System completes score calculation	System generates a result record.
Result is released	System makes the result available to the student.
Student selects View Result	System retrieves and displays the student's result.
Administrator selects Results	System displays authorized examination result information.

4.6.3 Functional Requirements

- **REQ-44:** The system shall generate a result after successful evaluation of an examination.
- **REQ-45:** The system shall store the result associated with the student and examination.
- **REQ-46:** The system shall calculate and display the percentage based on marks obtained and total marks.
- **REQ-47:** The system shall allow students to view their own results after the result is released.
- **REQ-48:** The system shall allow authorized administrators to view examination results.
- **REQ-49:** The system shall display relevant result information including examination name, marks obtained, total marks, percentage, and status where applicable.
- **REQ-50:** The system shall prevent students from viewing another student's result.

5 Other Nonfunctional Requirements

This section specifies the nonfunctional requirements of the Online Examination System, including performance, safety, security, software quality attributes, and business rules.

5.1 Performance Requirements

The following performance requirements apply to the Online Examination System:

- **PERF-01:** The system shall display the student dashboard within 3 seconds under normal operating conditions.
- **PERF-02:** The system shall display examination questions within 3 seconds after a valid request under normal operating conditions.
- **PERF-03:** The system shall process a valid answer submission within 2 seconds under normal operating conditions.
- **PERF-04:** The system shall process final examination submission within 3 seconds under normal operating conditions.
- **PERF-05:** The system shall generate a standard examination result within 5 seconds after evaluation for a normal-sized examination.
- **PERF-06:** The system should support at least 200 concurrent users without significant degradation in response time.

5.2 Safety Requirements

The Online Examination System does not directly control physical equipment and therefore has no significant physical safety requirements. However, the system shall include safeguards to prevent accidental data loss and destructive operations.

- **SAFE-01:** The system shall require confirmation before permanently deleting examination questions or examination records where deletion is supported.
- **SAFE-02:** The system shall prevent accidental modification of submitted examination answers.
- **SAFE-03:** The system shall maintain appropriate database backups to support recovery from data loss.

- **SAFE-04:** The system shall prevent an examination result from being generated from incomplete evaluation data.
- **SAFE-05:** The system shall preserve submitted answers when the examination is successfully submitted.

5.3 Security Requirements

The following security requirements shall be implemented to protect the Online Examination System and its data.

- **SEC-01:** The system shall require authentication before allowing users to access protected functions.
- **SEC-02:** The system shall implement role-based access control for students and administrators.
- **SEC-03:** The system shall prevent students from accessing administrator functions.
- **SEC-04:** The system shall allow only authorized administrators to create, update, or delete examination questions.
- **SEC-05:** The system shall allow only authorized administrators to create, update, schedule, or deactivate examinations.
- **SEC-06:** The system shall protect user credentials during transmission using secure communication mechanisms.
- **SEC-07:** The system shall not store user passwords in plain text.
- **SEC-08:** The system shall prevent students from accessing correct answers before the appropriate result or review stage.
- **SEC-09:** The system shall prevent a student from viewing another student's result.
- **SEC-10:** The system shall terminate inactive authenticated sessions after a predefined period where applicable.

5.4 Software Quality Attributes

The Online Examination System shall satisfy the following software quality attributes:

- **Usability:** The system shall provide a simple and consistent navigation structure across all major screens.

- **Reliability:** The system shall maintain examination, answer, and result data without loss during normal operation.
- **Availability:** The system should be available during scheduled examination periods except during planned maintenance or infrastructure failure.
- **Maintainability:** The system shall use a modular structure so that individual components can be modified without significantly affecting unrelated components.
- **Portability:** The web application shall be accessible through commonly used modern web browsers on supported operating systems.
- **Testability:** Each major system feature shall be independently testable using defined functional requirements.
- **Correctness:** The system shall calculate and store examination scores according to the configured marking rules.
- **Scalability:** The system architecture should allow expansion in the number of students, examinations, questions, and concurrent users.

5.5 Business Rules

The following business rules define the operating policies of the Online Examination System:

- **BR-01:** Only authenticated students shall be permitted to attempt examinations.
- **BR-02:** A student shall not be permitted to start an examination before its configured start time.
- **BR-03:** A student shall not be permitted to start an examination after its configured access period has ended.
- **BR-04:** An examination shall have a configured duration before it can be made available to students.
- **BR-05:** Only authorized administrators shall be permitted to create, update, or remove examination questions.
- **BR-06:** Only authorized administrators shall be permitted to create or schedule examinations.
- **BR-07:** A student shall be allowed to view only their own result.

- **BR-08:** Objective questions shall be evaluated using the correct answers and marking scheme configured by the administrator.
- **BR-09:** Once an examination is finally submitted, the student shall not be permitted to modify the submitted answers.

6 Other Requirements

The following additional requirements apply to the Online Examination System and are not covered in the previous sections.

- The system shall maintain examination records, submitted answers, and results for an appropriate period defined by the institution.
- The system shall maintain regular database backups to support recovery in the event of data loss.
- The system shall support English as the primary language of the user interface.
- The system shall maintain consistent date and time handling for examination scheduling.
- The system shall maintain an audit trail for important administrative operations where applicable.
- The system should allow future integration of additional question types without requiring a complete redesign of the system.

A Glossary

Term	Definition
OES	Online Examination System.
SRS	Software Requirements Specification.
Admin	Authorized administrator responsible for managing examinations and system operations.
Student	User who participates in online examinations.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
MCQ	Multiple Choice Question.

Authentication	Process of verifying the identity of a user.
Authorization	Process of determining which functions a user is permitted to access.
Question Bank	Collection of questions maintained for use in examinations.
Evaluation	Process of comparing submitted answers with correct answers and assigning marks.
Result	Examination performance information generated after evaluation.

B Analysis Models

This appendix contains analysis models used to represent different aspects of the Online Examination System.

B.1 Use Case Model

The major actors and their interactions with the Online Examination System include:

- **Student** – Login, View Available Exams, Start Exam, Answer Questions, Submit Exam, and View Result.
- **Administrator** – Login, Manage Users, Manage Questions, Create Exam, Schedule Exam, Manage Results, and View Reports.

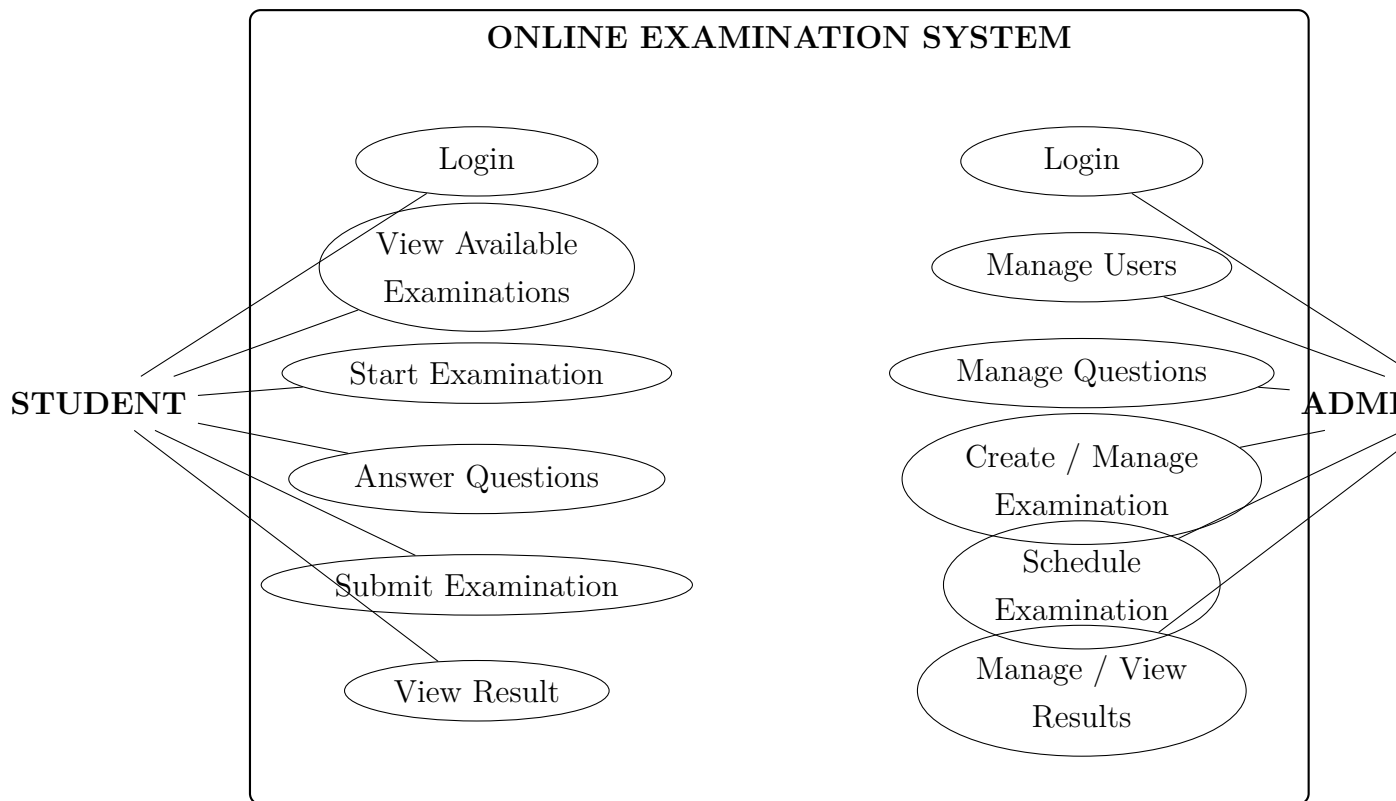


Figure 2: Use Case Model of the Online Examination System

B.2 Data Flow Model

The major data flow of the Online Examination System consists of user requests being processed by the application and stored or retrieved from the examination database.

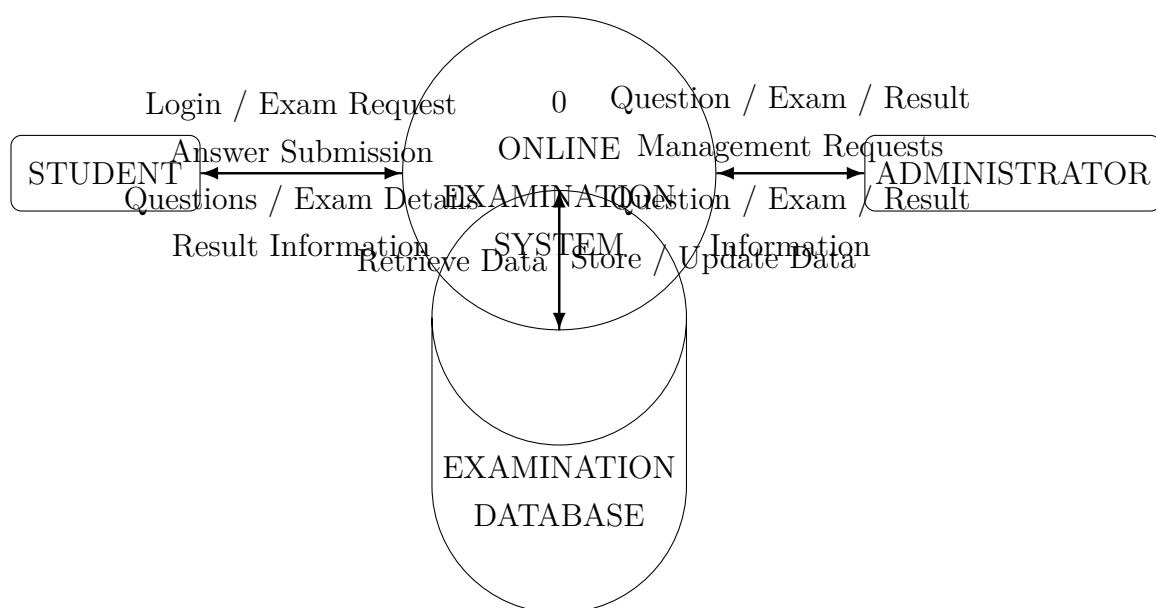


Figure 3: Data Flow Model of the Online Examination System

B.3 Entity-Relationship Model

The major entities of the Online Examination System include User, Examination, Question, Option, Answer, and Result.

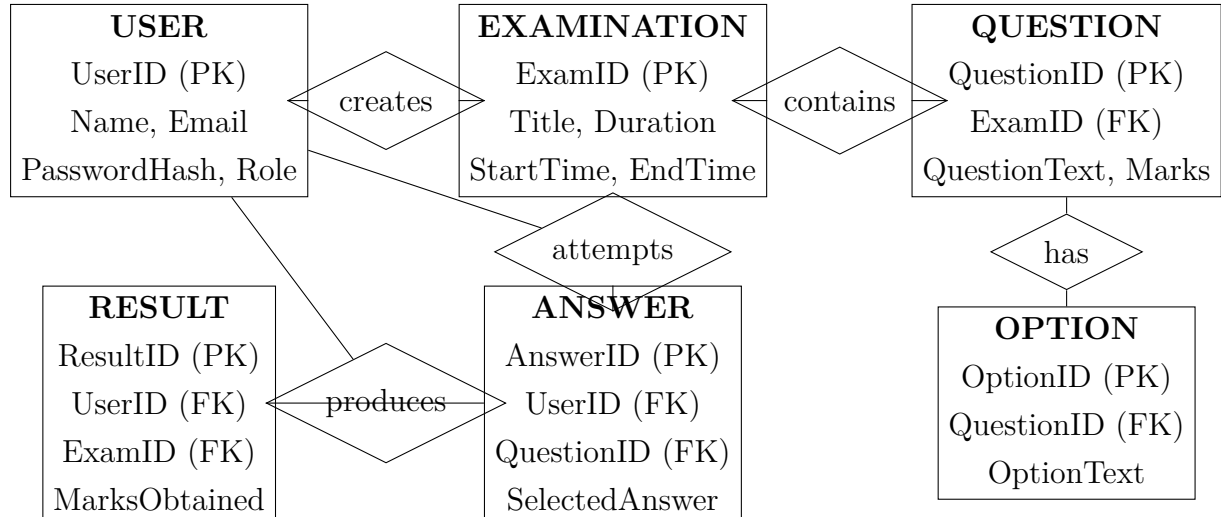


Figure 4: Entity-Relationship Model of the Online Examination System

C To Be Determined List

The following items have not yet been finalized and may be resolved during subsequent stages of the project.

Table 8: To Be Determined Items

TBD ID	Item	Status
TBD-01	Final maximum number of concurrent students during an examination	Pending
TBD-02	Final database backup frequency	Pending
TBD-03	Final result release policy	Pending
TBD-04	Final marking scheme, including any negative marking	Pending
TBD-05	Additional question types beyond objective questions	Pending